

Working of JS

JS is dynamically typed, cross platform threaded scripting and programming language. It is an asynchronous and concurrent language that offers a lot of flexibility.

↳ Execution context

Variable Env	Thread of execution
memory	Code
Key: value i.e;	o _____
a: 10	o _____
fn: {fn}	o _____
b: 1.2	o _____
x: true	o _____
y: "Hinu"	o _____

In thread of execution that is executed one line at a time.

In memory component which is an env. in CPU memory where variables are present in the form of key, value pairs & fun.

is present i.e., whole function is present.

- Single - Threaded means JS executes one command at a time.
- Synchronous - It only executes one command at a time in a specific order. i.e. we can go through the next line of code once the current lines has been completely executed.

(*) Note - we can make JS to work or behave asynchronously by using various block of code like! `async/await`, `promise` (covered earlier) etc.

↳ What happens when we run JS code.
"Every thing in JS happens inside the execution context"

```
function square (num) {  
    var ans = num * num;  
    return ans;  
}
```

```
var squared = square (2);  
console.log (squared);
```


↳ When we execute the code or any program, an execution context is created i.e., aka memory creation and code component creation phase.